



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



Exp :7 Write the Python Program to implement BFS

Aim:

To write a python program to implement BFS.

Algorithm:

- Step 1: Initialize a queue data structure and a set to keep track of visited nodes.
- Step 2: Add the starting node to the queue and to the set of visited nodes.
- Step 3: While the queue is not empty, dequeue the next node from the queue.
- Step 4: For each neighbor of the dequeued node, if it has not been visited yet, add it to the queue and mark it as visited.
- Step 5: Repeat steps 3-4 until the queue is empty.

Program:

```
from collections import deque
```

```
def bfs_traversal(graph, start_node):
```

```
    visited = set()
```

```
    queue = deque([start_node])
```

```
    visited.add(start_node)
```

```
    traversal_order = []
```

```
    while queue:
```

```
        current_node = queue.popleft()
```

```
        traversal_order.append(current_node)
```

```
    for neighbor in graph[current_node]:
```

```
        if neighbor not in visited:
```

```
            visited.add(neighbor)
```

```
            queue.append(neighbor)
```